MOHAMMAD TAAN AL AKOUM

AI & Machine Learning Engineer | MSc Artificial Intelligence, CentraleSupélec (Université Paris-Saclay)

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Availability: 6-month end-of-studies internship (stage de fin d'études) from April/May 2027; mobile within France and the EU

PROFILE

AI/ML engineer with a mechatronics background (graduated with distinction), completing an MSc in Artificial Intelligence at CentraleSupélec. I build trustworthy AI end to end, from multimodal predictive-maintenance models with calibrated uncertainty to production LLM systems with verifiable answers, as tested, reproducible code. Seeking an internship in industrial AI, LLM/RAG engineering or autonomous systems.

EDUCATION

MSc in Artificial Intelligence - CentraleSupélec, Université Paris-Saclay, France **2026 - 2027 (expected)**

Machine learning, deep learning, computer vision, NLP and LLMs, reinforcement learning, MLOps, predictive maintenance, AI planning.

BEng in Mechatronics Engineering - Rafik Hariri University, Mechref, Lebanon **2021 - 2025**

Graduated with distinction, GPA 85.65/100 (French equivalent: mention Très Bien); Dean's and President's Honour Lists.

RESEARCH EXPERIENCE

MSc Research Project - Multimodal Foundation Models for Predictive Maintenance **Oct 2026 - Present**

Laboratoire Génie Industriel (LGI), CentraleSupélec, Gif-sur-Yvette, France - supervised by Zhiguo Zeng

- **Architecture:** frozen encoders (sensor foundation model, DINOv2, Qwen3-Embedding) with evidential experts for calibrated uncertainty; fusion chosen by nested cross-validation on public bearing and milling datasets (CWRU, XJTU-SY, Paderborn, Nonastreda, MATWI), with KAIST and PHM 2010 held out as unseen machines.
- **Reproducibility:** reimplemented OmniFuser (arXiv 2025) as a tested PyTorch package and audited 35 paper-to-code items; found 134 of 512 milling passes duplicated and a label shortcut that beats the published model; designed a leakage-free leave-one-tool-out protocol with baselines, paired Wilcoxon tests and bootstrap intervals.
- **RUL-Fuse (remaining useful life of milling tools, forces + photos):** defined a causal target and horizon; a leakage probe excluded 2 of 3 photo types before any result; showed that OmniFuser with an RUL head (30 A100 fits) over-fits ten tools and loses to a simple usage rule (error 587 s vs 468 s), the bar for every gated design decision; test set sealed until the design freeze.
- **Engineering:** 512 automated tests, GitHub Actions CI (Ubuntu, Windows; Python 3.11, 3.12), Ruff, Pyright; 80+ A100 training runs as resumable SLURM arrays recording every configuration, seed and data checksum.

Code: github.com/taanakoum/Multimodal-AI-for-Predictive-Maintenance | /OmniFuser | /RUL-Model

PROJECTS

Agentic RAG Platform - multi-user document Q&A with verified citations, built solo **Autumn 2026**

- FastAPI and LangGraph agent streaming over SSE; Qdrant hybrid search (dense + BM25, reciprocal rank fusion, cross-encoder rerank) with an owner filter on every query; every citation verified server-side against the passages retrieved in that turn.
- Production patterns: PostgreSQL as source of truth, transactional outbox to Celery/Redis, idempotent jobs, sandboxed Docling parsing, per-conversation locks; Argon2id, refresh-token rotation, 2FA, OpenTelemetry.
- 1,193 automated tests on real PostgreSQL, Redis and Qdrant (93% line coverage); evaluated on real public documents (hit@1 0.64, against 0.94 on synthetic data); a real-browser test on an 80-page standard caught two bugs the unit tests missed.

Autonomous Smart Walker for Visually Impaired Users - final-year capstone, Rafik Hariri University **2024 - 2025**

- Designed and built end to end (frame, power, motor control, ROS on Raspberry Pi with Arduino): lidar SLAM, AMCL localisation, path planning with obstacle avoidance and Whisper voice commands; >90% indoor navigation success, 0.8-1.2 s voice to motion.

PROFESSIONAL EXPERIENCE

Reliability & Controls Engineering Intern - Ciment de Sibline, Sibline, Lebanon **Aug 2025 - Oct 2025**

- Programmed and validated PLC ladder logic (TIA Portal) for the cooling sequence and gear-rim lubrication of a critical ball mill; trended SCADA signals against alarm thresholds to find early warnings of lubrication, cooling and drive faults.

Automotive Diagnostics Intern - Bassoul & Hneine (BMW, Renault), Beirut, Lebanon **May 2025 - Jun 2025**

- Diagnosed vehicle faults by correlating diagnostic trouble codes with live ECU sensor data.

SKILLS

Machine Learning & AI: Python, PyTorch, scikit-learn, NumPy, Pandas, OpenCV; multimodal fusion, time series, anomaly detection, RUL prediction, foundation models (DINOv2, MOMENT, PatchTST), uncertainty, explainable AI.

LLMs & RAG: LangGraph, LangChain, Docling, Qdrant, cross-encoder rerankers, Hugging Face, Sentence Transformers, Pydantic structured output.

Backend & MLOps: FastAPI, PostgreSQL, SQLAlchemy, Redis, Celery, Streamlit, OpenTelemetry; SLURM, NVIDIA A100, Linux, Git, GitHub Actions, pytest, Playwright, Ruff, Pyright.

Robotics & Industrial: ROS, lidar SLAM, C++, Arduino, Raspberry Pi, MATLAB; PLC (TIA Portal), SCADA, condition monitoring.

Languages: English - C2 (IELTS); Arabic - native; French - A2, working towards B2 by the start of the internship.